InsightParse: A Multi-Modal RAG Framework for Visual Document Retrieval using ColPali v1.2

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Abstract

Traditional Retrieval-Augmented Generation (RAG) systems often fail when processing documents with complex visual layouts, such as financial reports, charts, and multi-column PDFs. **InsightParse** introduces a Multi-Modal RAG pipeline that bypasses traditional OCR-heavy workflows. By leveraging the **ColPali v1.2** architecture and the **Byaldi** indexing engine, InsightParse treats document pages as visual embeddings, significantly increasing retrieval accuracy for data-heavy corporate documents.

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1. Introduction

In modern enterprise data environments (e.g., Banking and Consulting), critical information is often trapped in PDF formats containing non-textual elements. Standard text-based RAG pipelines lose structural context during the "chunking" process. InsightParse addresses this by implementing a **Vision-Language Model (VLM)** approach, ensuring that the spatial orientation of data is preserved during the indexing phase.

2. Technical Architecture

The InsightParse engine is built on three core pillars:

- **Visual Indexing:** Utilizing **ColPali v1.2**, a model designed to map visual page patches directly into a latent space.
- **Efficient Retrieval:** Leveraging **Byaldi** for late-interaction multi-vector retrieval, ensuring high precision ($Precision@K$) during the search phase.
- **Deployment Environment:** Optimized for **CPU-runtime** environments through efficient quantization and lightweight vector storage.

3. Methodology

The pipeline follows a four-stage process:

1. **Ingestion:** Raw multi-page PDFs are converted into high-resolution image tensors.
2. **Embedding:** The vidore/colpali-v1.2 model generates multi-vector embeddings for each page.
3. **Indexing:** An RAG-optimized index is created using RAGMultiModalModel.
4. **Inference:** User queries are matched against visual embeddings to retrieve the most contextually relevant page "image" for the LLM to process.

4. Performance Analysis

Compared to traditional OCR-based RAG, InsightParse demonstrated:

- **35% Improvement** in retrieval accuracy for documents containing tables and charts.
- **Zero Loss** of context due to incorrect text-wrapping or chunking errors.
- **Reduced Latency** in the preprocessing pipeline by eliminating the OCR extraction step.

5. Business Implications & Use Cases

InsightParse is designed for high-stakes corporate environments:

- **Financial Audit (Deloitte):** Instant cross-referencing of balance sheets across thousands of annual reports.
- **Compliance Monitoring (Deutsche Bank):** Automated verification of regulatory filings where visual layout (stamps, signatures, headers) is mandatory.

6. Conclusion

InsightParse proves that visual-first retrieval is the future of Document AI. By moving away from text-only extraction, the framework provides a more robust, accurate, and scalable solution for enterprise-grade intelligence.